

YUZHONG CHEN

Phone: +86 18955797102 | Email: yuzhongchen2026@163.com

Personal Website: <https://russell12138.github.io>

EDUCATION

➤ University of Liverpool (UoL)

Liverpool, UK

BSc Hons in Computer Science

Sep. 2022 - May. 2026

- GPA: 3.88/4.0 | Major GPA: 4.0/4.0 | Rank: 1/214
- **Award:** XJTLU Excellence Scholarship (£7,000 per year), Computer Science Departmental Prize, Dean's List
- **Fellowship:** EPSRC-funded Summer Research Assistantship, UoL, 2025
- **Test Scores:** TOEFL 109 (R28, L29, S28, W24), GRE 329 (V159, Q170, AW3.5)

PAPERS

- [1] **Y. Chen**, Y. Qu, and H. Pu, "An analysis of attention mechanisms and its variance in transformer," *Applied and Computational Engineering*, vol. 47, no. 1, pp. 164-176, Mar. 2024, doi: 10.54254/2755-2721/47/20241291. (first author)
- [2] **Y. Chen**, T. Tan, A. Antala, S. Salaria, T. Saha, and Q. Tang, "Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs," submitted to *AAAI 2027*, under review. (first author)
- [3] **Y. Chen**, B. Konev, "Modular Expert Routing System for Large Language Models," undergraduate final year project.
- [4] **Y. Chen**, "RG-TAIL: Reliability-Gated Trust-Aware Imagination Learning under Limited Budgets," to be submitted to *ICLR 2027*. (single author)

RESEARCH PROJECTS

➤ Mixture-of-Encoders: Dynamic Routing System

Shanghai, China

Supervisor: Prof. Danushka Bollegala, UoL

Jun. 2026 - Present

- Proposed Mixture-of-Encoder, trained and distilled query-level universal router, which dynamically assign the most appropriate encoder based on the query embedding by an anchor encoder, along with a brief downstream task description

➤ RG-TAIL: Reliability-Gated Trust-Aware Imagination Learning

Shanghai, China

Self-proposed Independent Research

Apr. 2026 - Aug. 2026

- Conducted a large-scale empirical study (Real-or-Imagined, RoI) on synthetic-data utilization in model-based reinforcement learning, benchmarking real-only, imagined-only, and hybrid training across 6 environments, 3 RL algorithms, 6 data budgets, and 10 seeds to characterize when imagined experience is beneficial under data scarcity
- Found that world models generally outperform probabilistic world models in imagined and hybrid settings, suggesting that uncertainty estimates do not necessarily reflect the long-term utility of synthetic transitions for policy optimization
- Proposed RG-TAIL, a framework that estimates transition-level trustworthiness from held-out real-world prediction errors, and introduces a reliability-gating mechanism based on world-model agreement and rollout consistency to adaptively determine whether synthetic experience should participate in policy optimization
- Evaluated RG-TAIL against both internal baseline RoI, and SOTA model-based RL methods (MBPO, TD-MPC2, DreamerV3), demonstrating superior average performance and task success rates under low-budget settings across diverse control environments and RL algorithms; further conducted extensive mechanistic analysis to diagnose gating behavior, identify failure modes, and validate the effectiveness of the proposed design

➤ MERS: Modular Expert Routing System for Large Language Models

Liverpool, UK

Supervisor: Prof. Boris Konev, Dean of the School of Computer Science and Informatics, UoL

Sep. 2025 - May. 2026

- Designed and implemented MERS, a modular architecture built from multiple domain-specialized expert models coordinated by a classifier-based routing mechanism, inspired by RAG and Mixture-of-Experts principles
- Initialized two baselines: unified (single LLM fine-tuned on full corpus) and modular (per-domain experts along with classifier routing) on 10 disciplinary subsets from MMLU, and benchmarked MERS for feasibility verification
- Conducted two core evaluation studies: (1) Efficiency comparison by quantifying performance, inference latency, and token-costs under unified vs. modular setups; (2) Generalization assessment by evaluating cross-domain robustness on 10 unseen MMLU subsets and characterizing the specialization-generalization trade-off with empirical evidence

➤ **Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs** **Liverpool, UK**

Supervisors: Asst. Prof. Tulika Saha, UoL and Asst. Prof. Qiyi Tang, UoL

Oct. 2024 - Jan. 2026

- Proposed MOCA-MLLM, a modality order-aware controlled architecture for MLLMs, with a unified preprocessing pipeline for twelve multimodal benchmark datasets (MELD, CMU-MOSI, HateMM, RLTD, etc.), enabling order-wise injection of text, vision, and audio embeddings at predefined decoder layers for large-scale, controlled ablation studies
- Conducted extensive fine-tuning experiments across four MLLMs, six downstream tasks, and six injection sequences, revealing consistent, task-dependent modality order preferences: affective recognition tasks (sentiment, emotion) benefit from early visual/acoustic cues, whereas semantic-grounded tasks (sarcasm, humor, hate speech, deception) consistently favor early access to text, with text-middle configurations emerging as the most consistently suboptimal
- Demonstrated that carefully designed sequential modality injection consistently outperforms All-at-Input baselines, showing that increased processing depth alone does not guarantee improved performance

➤ **An Analysis of Attention Mechanisms and Its Variance in Transformer** **Beijing, China**

Supervisor: Assoc. Prof. Zengchang Qin, Beihang University

Jun. - Sep. 2023

- Implemented and evaluated a suite of ML algorithms (DT, KNN, NB) and neural architectures including CNNs for license plate and traffic sign recognition, and RNNs (LSTM, BiLSTM) for sentiment and emotion classification
- Investigated Transformer variants, focusing on linearization, sparsification, multi-head self-attention and low-rank decomposition; conducted ablation studies to verify the performance improvement and highlight the trade-offs between scalability, interpretability and computational performance in Transformer-based architectures

PROFESSIONAL EXPERIENCE

➤ **Shanghai Institute of Computing Technology** **Shanghai, China**

Research & Development Center Intern

Jul. - Sep. 2024

- Engineered and deployed a client-facing multimodal chatbot interface using Gradio with streaming inference and asynchronous I/O; supported text, image, file, and table inputs to demonstrate real-time reasoning and model capabilities
- Constructed a domain-specific agricultural knowledge base with LangChain and Word2Vec; optimized InternLM's RAG pipeline by integrating curated Q&A datasets and custom retrievers, improving top-k retrieval precision by 18%
- Developed a multi-turn real-time weather assistant agent, combining OpenWeatherMap API, prompt engineering and dynamic state tracking; conducted ablation studies on prompt fidelity, hallucination control and dialogue continuity

COURSEWORK PROJECTS

➤ **AI-Powered Recruitment Management System Development** **Jan. - May. 2025**

- Built a multi-user recruitment platform with candidate, recruiter, and admin modules; integrated OpenAI API and RAG for intelligent resume-job matching, and deployed backend via Flask and Alibaba Cloud with real-time chat functionality

➤ **Simulation and Optimization of Leader-Election Algorithms** **Jan. - Mar. 2025**

- Implemented distributed algorithms (LCR, HS) in Java; verified theoretical complexity bounds ($O(n)$, $O(n \log n)$) and reduced communication cost by up to 60% through algorithmic optimization and large-scale network simulations

➤ **Adversarial Attack and Defense in Neural Networks** **Sep. - Dec. 2024**

- Implemented framework (FGSM, PGD, adversarial training) on CNN baseline, ranking top 5% in robustness evaluation

➤ **Python-Based Data Visualization and Feature Analysis** **Mar. - May. 2024**

- Built a unified Python ML pipeline integrating feature extraction, dimensionality reduction (PCA, t-SNE, LDA), and both supervised and unsupervised learning; optimized classifier ensemble achieving top 5% performance among peers

ADDITIONAL

- **Programming Languages:** Python (4 yrs), Java (3 yrs), SQL (3 yrs), C (2 yrs), JavaScript (1 yr), HTML (1 yr)
- **Frameworks & Tools:** GitHub, Hugging Face, Jupyter, Overleaf, MATLAB, MySQL, Linux, Google Cloud Platform